

AMAZON ML CHALLENGE 2026

Business Entity Resolution

Hackathon Project Documentation & 72-Hour MVP Execution Plan

| Participant     | Details                                       |
|-----------------|-----------------------------------------------|
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| Career Identity | Machine Learning Engineer                     |
| Education       | B.Tech CSE — 4th Year                         |
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| Challenge       | Amazon ML Challenge 2026                      |
| Focus           | Business Entity Resolution                    |

PROJECT MISSION

Build a reproducible, competition-grade MVP that maximizes entity-level macro  $F_{0.5}$  through high-recall candidate generation, strong pairwise features, precision-focused matching, threshold optimization and disciplined validation.

Challenge Window: 25 Sep 2k26, 12:00 AM IST → 27 Sep 2k26, 11:59 PM IST

## 1. Document Control

| Item             | Value                                            |
|------------------|--------------------------------------------------|
| Version          | v1.0 — Hackathon Execution Documentation         |
| Prepared By      | Sunil Kumar                                      |
| Role             | Machine Learning Engineer                        |
| Date Prepared    | 26 September 2026                                |
| Challenge Period | 25–27 September 2026                             |
| Primary Metric   | Entity-level Macro $F_{0.5}$                     |
| Submission Limit | Maximum 5 submissions per day                    |
| Source Basis     | Official problem statement + official guidelines |

## 2. Executive Summary

The Amazon ML Challenge 2026 Business Entity Resolution problem requires matching records from Source 2 and Source 3 to each Source 1 business. Records contain noisy and inconsistent names, addresses and country labels; a Source 1 entity may have zero, one or multiple matches.

The proposed solution is retrieval-first: normalize records, generate high-recall candidates through multi-pass blocking, build pairwise similarity features, train a supervised matcher, and optimize the final decision threshold for entity-level macro  $F_{0.5}$ . The MVP prioritizes measurable performance, speed, reproducibility and valid submission generation.

### CORE PRINCIPLE

Blocking creates the recall ceiling. Matching creates discrimination. Thresholding controls precision. Validation decides which changes are useful.

## 3. Problem Definition

### 3.1 Business Entity Resolution

- Source 1 is the deduplicated reference source.
- Find all matching Source 2/Source 3 records for every Source 1 entity.
- Zero, one or many matches are valid.
- Business names and addresses contain realistic noise.
- All files and submissions use TSV format.

### 3.2 Data Schema

| Field            | Meaning                                      | Use                              |
|------------------|----------------------------------------------|----------------------------------|
| entity_id        | Unique ID; S1/S2/S3 prefix identifies source | Routing and validation           |
| business_name    | Business name                                | Primary textual signal           |
| business_address | Business address                             | Secondary/discriminative signal  |
| country          | Country label                                | Open-set feature/blocking signal |

### 3.3 Noise Patterns

- Abbreviations and legal suffix changes.
- Punctuation, word-order changes and typos.
- DBA/trade names and transliteration.
- Address abbreviations and missing components.
- Landmarks, numbering variations and reordering.

## 4. Evaluation & Rules

### 4.1 Metric

The challenge uses entity-level macro  $F_{0.5}$ . Because  $F_{0.5}$  weights precision more heavily than recall, false matches must be controlled without allowing blocking recall to collapse.

| Concept          | Project Rule                               |
|------------------|--------------------------------------------|
| Precision        | Avoid unrelated matches.                   |
| Recall           | Recover true matches.                      |
| $F_{0.5}$        | Primary optimization target.               |
| No-match         | Do not force a match.                      |
| Multiple matches | Allowed; do not force one-to-one matching. |

### 4.2 Official Constraints

- Challenge window: 25–27 September 2026, 12:00 AM IST to 11:59 PM IST.
- Maximum 5 submissions per day for 3 days.
- Public and Private leaderboards are used; final leaderboard is revealed after the challenge.
- Best solution artifacts include a 1–2 page ML approach document and commented source code.
- Top 100 teams may need to provide methodology, blocking/candidate generation, model architecture and feature engineering.
- Desktop/laptop only; simultaneous logins are not allowed.
- Cheating, plagiarism, multiple IDs and unfair practices can lead to disqualification.

### 4.3 Data Restrictions

The project will not use external business registries, search-engine lookups, maps, geocoding, company websites or entity-enrichment services to resolve individual challenge records. General technical research is limited to algorithms and engineering methods.

## 5. Participant Profile & Objectives

Sunil Kumar is a 4th-year B.Tech CSE student at BVC Group of Institutions, Rajahmundry, India, with a primary career identity of Machine Learning Engineer. This hackathon project is designed to demonstrate practical machine learning, NLP-style record linkage, information retrieval, feature engineering, evaluation and reproducible ML engineering.

- Build a real end-to-end MVP.
- Understand the actual dataset before making assumptions.
- Develop high-recall blocking.
- Train a precision-focused matcher.
- Optimize entity-level  $F_{0.5}$ .
- Use leaderboard submissions strategically.

- Maintain experiment history and reproducibility.
- Produce a defensible final methodology.

## 6. Technical Architecture

| Stage          | Input                  | Output            | Purpose                                |
|----------------|------------------------|-------------------|----------------------------------------|
| Data Forensics | Raw TSV                | Profile           | Understand scale/schema/noise          |
| Normalization  | Names/addresses        | Normalized fields | Reduce superficial variation           |
| Blocking       | Normalized records     | Candidates        | Protect recall and reduce search space |
| Features       | Candidate pairs        | Feature matrix    | Quantify similarity                    |
| Matcher        | Features               | Match scores      | Separate true links/hard negatives     |
| Decision       | Scores                 | Accepted matches  | Optimize $F_{0.5}$                     |
| Output         | Predictions/candidates | 2 TSV files       | Create submission                      |
| Validation     | Outputs                | Report            | Check quality/format                   |

### PIPELINE

Raw TSV → Profiling → Normalization → Multi-pass Blocking → Final Candidate Set → Pairwise Features → ML Matcher → Threshold Optimization → Decision Layer → matching\_results.tsv + candidate\_pairs.tsv → Official Validator

## 7. Research & Model Strategy

### 7.1 Blocking

- Exact normalized-name blocks.
- Country + name-prefix blocks.
- Address token/component blocks.
- Name + address composite blocks.
- Character n-gram TF-IDF retrieval.
- Word/token TF-IDF retrieval.
- Efficient fuzzy retrieval.
- Data-driven blocks discovered from supplied records.

Every blocking experiment will measure true-match coverage, candidate volume, reduction ratio, runtime and memory. The target is high recall with manageable candidate volume.

### 7.2 Pairwise Features

- Exact and normalized equality.
- RapidFuzz ratio/partial/token-sort/token-set similarities.
- Character and word TF-IDF cosine similarity.
- Token Jaccard and shared-token ratios.
- Length/token-count differences.
- Numeric address overlap.
- Postal/city/state agreement where reliable.
- Country equality and missingness.
- Name × address interaction features.
- Candidate-generation source indicators.

## 7.3 Models

- Logistic Regression as an interpretable baseline.
- HistGradientBoostingClassifier or another compliant gradient-boosting model.
- Investigate other boosting libraries only when licensing/packaging is compatible.
- Prioritize hard-negative mining over unnecessary deep-learning complexity.
- Use calibration only if it improves measured entity-level  $F_{0.5}$ .

## 8. MVP Scope

- Automatic local dataset discovery and profiling.
- Memory-aware loading.
- Field-specific normalization.
- Multi-pass blocking and blocking-recall measurement.
- Pairwise feature generation.
- Supervised matcher.
- Entity-level macro  $F_{0.5}$  evaluation.
- Threshold optimization.
- Singleton/no-match protection.
- Hard-negative sampling/mining.
- Test inference.
- matching\_results.tsv and candidate\_pairs.tsv.
- Official validator check.
- Experiment tracking and reproducible configuration.

### 8.1 Not in First MVP

- LLM as the primary matcher.
- External business-data enrichment.
- Full Cartesian matching.
- Unvalidated graph/deep-learning architecture.
- Complex UI that does not improve the model.
- Manual test-set labeling.

## 9. 72-Hour Execution Timeline

### 9. 72-Hour Execution Timeline

| Time              | Phase          | Key Actions                                                                       | Deliverable            |
|-------------------|----------------|-----------------------------------------------------------------------------------|------------------------|
| Day 1 — Early     | Forensics      | Inspect files, schema, counts, missingness, countries, ground truth and hardware. | dataset_profile.md     |
| Day 1 — Morning   | Normalization  | Build/test name and address normalization.                                        | Normalization module   |
| Day 1 — Afternoon | Blocking       | Exact/prefix/address/composite blocks; measure recall.                            | Blocking report        |
| Day 1 — Evening   | Training Data  | Generate candidates + realistic hard negatives.                                   | Candidate training set |
| Day 1 — Night     | Baseline Model | Features + baseline matcher + local $F_{0.5}$ .                                   | Baseline model         |
| Day 2 — Morning   | Features v2    | Fuzzy, TF-IDF, token, numeric and interaction features.                           | Feature v2             |
| Day 2 — Afternoon | Hard Negatives | Mine high-scoring false positives; retrain.                                       | Improved model         |
| Day 2 — Evening   | Threshold      | Search threshold and analyze singleton/ambiguity errors.                          | Threshold report       |

|                   |                 |                                                   |                    |
|-------------------|-----------------|---------------------------------------------------|--------------------|
| Day 2 — Night     | Submission      | Validate and submit strongest configuration.      | Submission version |
| Day 3 — Morning   | Error Analysis  | FP/FN/common-name/country/missing-field analysis. | Error report       |
| Day 3 — Midday    | Blocking v2     | Add only validated retrieval improvements.        | Blocking v2        |
| Day 3 — Afternoon | Final Model     | Freeze strongest validated configuration.         | Final model        |
| Day 3 — Evening   | Final Inference | Generate both TSVs; run official validator.       | Final outputs      |
| Day 3 — Final     | Documentation   | Freeze code, history, methodology and package.    | Submission package |

### SUBMISSION DISCIPLINE

There are up to 5 submissions per day. Each submission should test a documented hypothesis and be preceded by local validation plus the official validator.

## 10. Submission & Experiment Strategy

| Day   | Submission Focus                                                                     | Purpose                                         |
|-------|--------------------------------------------------------------------------------------|-------------------------------------------------|
| Day 1 | Baseline → blocking → features → threshold → best Day-1                              | Establish and improve the first working system. |
| Day 2 | Hard negatives → advanced features → retrieval combinations → decision logic         | Improve discrimination and precision.           |
| Day 3 | Error fixes → conservative variant → recall variant → final candidates → final model | Select a robust final configuration.            |

Maintain an experiment CSV containing experiment ID, blocking strategy, blocking recall, candidate count, features, negative sampling, model, threshold, local precision/recall/ $F_{0.5}$ , runtime, memory, leaderboard score and notes.

## 11. Validation Framework

- Hold out validation data from training.
- Evaluate at Source 1 entity level, not only pair level.
- Calculate precision, recall and  $F_{0.5}$  per entity and macro-average.
- Search thresholds rather than assuming 0.5.
- Treat empty true/predicted match sets explicitly.
- Do not force one-to-one matching.
- Inspect false positives, false negatives and ambiguous candidates.

### 11.1 Error Categories

- Singleton false positives.
- Missed true matches.
- Ambiguous top-two candidates.
- Common-name collisions.
- Name-only false positives.
- Address-only false positives.
- Missing-field cases.
- Cross-source errors.
- Country-specific formatting errors.

## 12. Final Output Specification

### 12.1 matching\_results.tsv

| source1_entity_id | matched_entity_ids |
|-------------------|--------------------|
| S1-00001          | S2-00047,S3-00812  |
| S1-00002          | S3-00004           |
| S1-00003          |                    |

### 12.2 candidate\_pairs.tsv

| source1_entity_id | candidate_entity_ids       |
|-------------------|----------------------------|
| S1-00001          | S2-00047,S2-00193,S3-00812 |
| S1-00002          | S3-00004,S3-00115          |
| S1-00003          |                            |

The candidate file must be the final candidate set immediately before model scoring. Every predicted match must exist in the corresponding candidate list.

## 13. Proposed Repository Structure

business-entity-resolution/

```

├── dataset/
├── src/business_entity_resolution/
│   ├── config.py
│   ├── data_loader.py
│   ├── profiling.py
│   ├── normalization.py
│   ├── blocking.py
│   ├── candidate_generation.py
│   ├── features.py
│   ├── negative_sampling.py
│   ├── model.py
│   ├── training.py
│   ├── validation.py
│   ├── metrics.py
│   ├── threshold.py
│   ├── inference.py
│   ├── postprocessing.py
│   ├── submission.py
│   └── pipeline.py
├── notebooks/
├── models/
├── cache/
├── experiments/
├── reports/
├── validation/
├── output/
├── tests/
├── requirements.txt
├── README.md
└── run_pipeline.py

```

## 14. Risk Management

| Risk                                | Impact   | Mitigation                                                    |
|-------------------------------------|----------|---------------------------------------------------------------|
| Missed true matches during blocking | High     | Multi-pass retrieval + blocking-recall measurement.           |
| Candidate explosion                 | High     | Indexed retrieval, top-K controls, validated filters.         |
| False positives                     | High     | Hard negatives, address features, threshold/ambiguity checks. |
| False negatives                     | High     | Improve retrieval recall and complementary blocks.            |
| Memory overflow                     | High     | Chunking, sparse matrices and memory-aware storage.           |
| Leaderboard overfitting             | High     | Local validation and private-leaderboard robustness.          |
| Invalid submission                  | High     | Official validator before submission.                         |
| Time lost to complexity             | High     | Freeze MVP before advanced experimentation.                   |
| Rule violation                      | Critical | Use only permitted data/tools and preserve version history.   |

## 15. Definition of Done

| Checkpoint                     | Target                            |
|--------------------------------|-----------------------------------|
| Dataset inspected/profiled     | Complete before advanced modeling |
| Normalization tested           | Complete                          |
| Blocking recall measured       | Complete                          |
| Candidate generation validated | Complete                          |
| Pairwise features              | Complete                          |
| Baseline matcher               | Complete                          |
| Entity-level $F_{0.5}$         | Complete                          |
| Threshold optimized            | Complete                          |
| Hard negatives investigated    | Complete                          |
| Error analysis                 | Complete                          |
| Test inference                 | Complete                          |
| Both TSV outputs               | Complete                          |
| Official validator             | Mandatory                         |
| Experiment history             | Mandatory                         |
| README/methodology             | Mandatory                         |
| Reproducible final code        | Mandatory                         |

## 16. Final Methodology Outline for Official Submission

1. Problem: Business Entity Resolution across three noisy sources.
2. Data: Source 1 reference records, Source 2/3 records, ground truth and unlabeled test set.
3. Preprocessing: Unicode-safe normalization and field-specific representations.
4. Candidate Generation: Multi-pass exact, prefix, address, TF-IDF and fuzzy retrieval with measured blocking recall.
5. Features: Name/address similarity, TF-IDF, token overlap, numeric address overlap, country and interaction features.
6. Model: Interpretable baseline followed by a validated gradient-boosting matcher if it improves macro  $F_{0.5}$ .
7. Training: Ground-truth positives plus hard negatives sampled from realistic candidate pools.
8. Validation: Entity-level macro  $F_{0.5}$  with explicit no-match/singleton handling.
9. Optimization: Threshold and ambiguity rules tuned only on held-out validation.
10. Output: matching\_results.tsv and candidate\_pairs.tsv, verified by the official validator.
11. Conclusion: Select the strongest validated and reproducible configuration.

## 17. Sunil Kumar — Personal Hackathon Checklist

| Priority | Task                             | Target            |
|----------|----------------------------------|-------------------|
| P0       | Understand dataset and rules     | Before model work |
| P0       | Working end-to-end MVP           | Day 1             |
| P0       | Blocking recall                  | Day 1             |
| P0       | First valid submission           | Day 1             |
| P1       | Feature improvements             | Day 2             |
| P1       | Hard-negative mining             | Day 2             |
| P1       | $F_{0.5}$ threshold optimization | Day 2             |
| P1       | Error analysis                   | Day 2–3           |
| P1       | Validated blocking improvements  | Day 3             |
| P0       | Final submission + documentation | Before deadline   |

### PERSONAL RULE

Do not chase complexity. Build the complete pipeline first, measure it, then improve the weakest measured stage.

## 18. Official Reference Materials

- Amazon ML Challenge 2026 — Business Entity Resolution Challenge Problem Statement.
- Amazon ML Challenge 2026 — Guidelines and Key Instructions.
- Official challenge validator and supplied challenge documentation in the project workspace.

This project document does not replace the official challenge rules. If the official platform or instructions are updated, the latest official instructions take precedence.

## 19. Revision History

| Version | Date              | Change                                                                                                                                                          |
|---------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.0    | 26 September 2026 | Initial professional hackathon documentation with participant profile, technical architecture, 72-hour timeline, validation, submission strategy and MVP scope. |

## AMAZON ML CHALLENGE 2026

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